KOMBI SHIFT 2

Service, competition and route expansion

Current design and engineering evidence - 10 November 2026

CONNECTED ITINERARY	SERVICE LOCATIONS	VALIDATION SCOPE
1.25 km	7 stops / 2 stages	Automated + platform checks; human testing remains



The existing Unity game is extended in place. This report separates implemented features, executed technical evidence and work that still requires people. It does not claim a complete rubric score or fabricated playtest results.

Unity 6000.6.2f1 | Branch feat/kombishift2-unity6 | Assessment deadline 13 November 2026

1 / Current design

Updated 9 November 2026, HAI311 Group 6, deadline 13 November 2026. This source description supersedes the 8 November design. The new expansion PDF under output/pdf describes this implementation; the earlier illustrated PDF, presentation, submission copies and historical build evidence describe the prior release. Human participation and permissions are not inferred from software changes.

Identity, audience and rationale

A Harare-inspired 3D kombi-driving game for casual keyboard/mouse players aged 14+, particularly students familiar with Zimbabwean transport. This audience is a design hypothesis. Short shifts, visible service requests, identifiable rivals and severity-based consequences make driving choices observable. The game is an arcade simulation, not driver training or a statement of statutory Zimbabwean fines.

The existing imported city, Hiace, passenger rigs, RCC integration, scene GUIDs and title/pause/results flow are retained. The expanded playable itinerary uses connected, previously unused bz0/bz1 roads before the original bz2 city route. It spans approximately 1.25 km and seven service locations. Reusing existing collision geometry and road assets avoids a second disconnected prototype. This is a connected bidirectional itinerary, not a general urban lane-graph pathfinder.

Player loop and progression

Select Morning Shift or Rush Hour on the title screen (F2), start, locate a queue, drive and stop below 2 km/h. Hold Space while passengers walk through the doorway and occupy separate seats. Each real seat arrival creates a PassengerTrip, with origin, downstream destination, fare, happiness, patience, state and request flag. Alternate arrivals choose the next valid service stop or Mbare; this deterministic mix guarantees intermediate and terminal trips without invalid upstream destinations.

Near a destination, passengers request a stop through visible text and a locally generated bell. Stop in its service radius to run the existing alighting character animation. A completed alighting trip pays its configured fare and earns service XP. Intermediate deliveries count toward the shift target; the player must finish at Mbare before time expires. Fares are earned on completed delivery, replacing the earlier boarding-income rule. A success time bonus is round(remaining seconds x 0.08). Best successful net earnings persist by stage.

A missed request is detected after approaching within 90 m and then receding beyond 144 m. Its one-shot penalty reduces that passenger's happiness by a default 20 and driver XP by 10. A missed passenger can alight at any serviced station, including Mbare. Destination requests are evaluated by one half-second manifest sampler, not a separate polling component per passenger. The minimap target prioritises requested destinations.

Morning: 600 seconds, target 1+, $2 per completed trip, 60% traffic, conservative/competitive rivals, 6 m/s base rival speed and 45 s player head start plus stagger. Rush Hour: 420 seconds, target 6+, $3 per completed trip, all traffic, aggressive/competitive rivals, 9 m/s base rival speed and 8 s head start plus stagger.

Assets/Resources/MorningShift.asset and RushHour.asset expose duration, target, rival speed/head start, traffic fraction and fare. Parameters are initial balance values, not conclusions from authentic testers. An expanded route increases travel cost while intermediate alighting increases seat reuse; measure completion and missed-stop rates before raising targets.

Rival intelligence and physics

RivalKombiAI retains Cruise, ChasePassenger, PullIn, Boarding and Depart. An explicit evaluation pass scores passenger demand capped by free seats, actual forward route distance/travel time, nearby congestion and reservation status. Already reserved queues are ineligible. Reassessment cooldown and a required utility advantage prevent oscillation. Exhausted queues trigger a new search; full rivals proceed to Mbare. Queue ownership still uses PassengerStation.TryReserve, and only physical seat arrivals increment rival totals.

Rivals now use dynamic rigidbodies and four runtime WheelColliders with suspension, bounded steering, motor torque and braking torque. Their root position is not teleported or advanced with MovePosition. The cloned visual vehicle remains intact. A low centre of mass, suspension substeps, continuous collision detection, acceleration limits and speed-dependent corner braking constrain the vehicle.

Left-lane offsets follow travel direction for both rivals and existing traffic. A forward-only path evaluation prevents choosing an upstream queue by simply reversing through traffic. At itinerary endpoints, vehicles must physically turn. This is not yet a junction-level lane graph with designated turning bays.

Conservative drivers keep longer gaps and do not overtake. Competitive drivers accept smaller gaps and safe passes. Aggressive drivers use higher speed, can deliberately pass in a no-overtaking zone and can roll through stop controls; they still require forward/rear/side clearance before passing. A pass changes its desired lane offset gradually; the wheels perform the steering. Clearances, stop signs, speed/hump zones and traffic signals influence speed. A blocked vehicle reassesses demand and waits for clearance; there is no teleport recovery or alternate-road graph search.

Rivals have distinct property-block body colours without modifying shared material assets. Names, totals, target distance and matching minimap colours identify competitors. Activity events announce target choices, pickups, departures and passing decisions. Aggressive illegal-passing decisions increment a rival statistic; not every player offence is yet mirrored into a full rival penalty record.

Happiness, XP, damage and economy

KombiServiceQuality broadcasts reaction text and XP changes from one vehicle sampler. Longitudinal/lateral acceleration detect harsh braking, acceleration and steering. Collision and traffic-law systems publish discrete penalties. Cooldowns prevent per-frame punishment. Smooth moving service can earn a small reward after 30 seconds; parking does not farm smooth-driving XP. Passenger happiness is bounded 0-100, and the HUD averages currently active trips. Completed records retain their final score. XP is a separate signed shift total.

KombiVehicleCondition uses collision relative velocity plus impulse/mass. Contacts below 2.5 m/s or weak impulse are ignored, damage is severity scaled, capped at 35 per incident and rate limited. Health bottoms at 10. The custom keyboard-to-RCC adapter multiplies requested throttle and braking by bounded condition factors. Lower power reduces acceleration; brake effectiveness remains at least 75%. The system does not modify vendor DLLs, deform meshes or detach wheels. Severe damage currently has HUD/reaction feedback; dedicated smoke, impact VFX and damaged-engine audio remain polish work.

T repairs at any station within its service radius, below 2 km/h, while the shift is active. Repair price is ceil(missing health x 0.15). It requires sufficient earned funds and restores health only after payment. Fines subtract net earnings and remain separately itemised; a negative shift balance is permitted.

Roads and traffic rules

The editor upgrade creates three additional queue locations on existing connected roads, extends player/minimap/rival/traffic route references, and adds reusable shallow mesh humps. The hump is 4 m long with a 12 cm peak and no high vertical edge. Warnings precede it. Crossing too fast causes a rate-limited complaint; severe overspeed can reduce condition. Physical response uses the actual road collider, hump mesh and RCC suspension. Driven hump validation is separately recorded; geometry alone is not proof of acceptable ride quality.

KombiRoadRuleZone registers configurable speed, stop, crossing, roadworks, no-entry, direction, no-overtaking, hump and give-way zones. Trigger bookkeeping handles multiple colliders per vehicle. Speed zones enforce the lowest active limit. Continuous overspeed has tolerance and a three-second grace period, then one severity-scaled fine until compliance. Stationary speed below the limit re-arms the next incident. Stop-zone exit checks whether a full stop occurred; no-entry triggers a fine, direction checks actual velocity, and prohibited passing checks directed position on the right lane. Incident keys and cooldowns suppress repeat charges.

Give Way is presently a warning/AI caution, not a reliable yielding-violation detector. No claim is made for offences that cannot be reliably detected. All monetary values are configurable game-design values. Additional alternate road branches, wider/narrower road construction and complete intersection right-of-way arbitration remain scope gaps.

UI, controls and architecture

WASD/arrows drive; Space parking brake; C camera; RMB orbit; wheel zoom; Escape pause; R restart; T repair. Existing menu keyboard focus, help, camera sensitivity/distance and master volume remain. The existing speed/minimap/objective layout is extended with secondary condition, speed-limit, XP, happiness, delivered and fine information. Stop requests have a separate alert area. Rival rows show proximity and totals. Results count cumulative completed trips rather than the current cabin count.

Director owns timer, progression, active manifest, completed trips and net earnings. Stations own exclusive queue reservations. StationPassenger owns physical character states. PassengerTrip owns trip state/data. ServiceQuality owns reaction cooldowns/XP. VehicleCondition owns health/repair. TrafficLaw owns offence incidents/records. RoadRuleZone associates rules with physical areas. RivalDrivingProfile holds exposed utility/driving parameters. StageSettings is reusable stage configuration. Fare, pause, shift-finished, trip-changed, rival-activity, quality and damage events connect feedback to rules.

Animation, assets and attribution

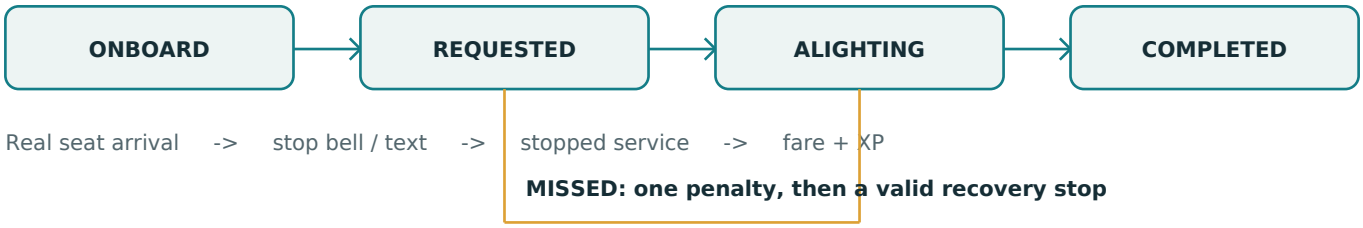
Imported humanoid rigs continue to use the existing procedural seated/walk/alight pose blending. This is not a newly authored Animator clip graph. RCC supplies player vehicle physics; custom code supplies integration, conditions and service rules. The request/complaint sound is a locally generated tone, with text reactions; it is not recorded speech. Existing assets and licence notices are retained. No paid dependency or external speech service was added. OpenAI Codex assisted implementation and verification at Jayden's request; members must record and understand actual work. Source redistribution permissions, including the RCC notice, still require resolution.

Evidence and honest limitations

See Docs/Validation/Expansion and EXPANSION_ENGINEERING_REPORT.md for executed test/build results. Historical 8 November evidence remains separately dated. Rule fixtures, staged integration, headless physics, exported builds, native execution and human driving are different evidence scopes. The first integration failures are retained rather than erased. An assertion can be corrected when the game deliberately changes its economic contract; unrelated failures must be investigated.

At least three authentic human sessions, severity analysis, three observation-linked fixes and same-task retests remain required. No tester, feedback, group contribution, assigned-book consultation or panel exercise is fabricated. Native Windows/weak-machine/browser usability and accessibility still need genuine human hardware evidence. Existing PDF/deck/submission snapshots must be regenerated from this updated source before hand-in.

2 / Passenger service flow



Character movement states remain separate from the trip data above. A fare is paid after a real alighting completion, and a missed request is penalised once. Rival queues use the same station ownership and physical boarding callbacks.

Rendered scene evidence



Actual Editor capture from a staged verification fixture. Background agents may be isolated; this is not a human playtest.

3 / Engineering report

10 November 2026. Unity 6000.6.2f1. Branch feat/kombishift2-unity6. Earlier releases and submission snapshots are preserved. This report describes substantial implementation work; it does not claim all 100 rubric marks or the complete requested definition of done.

Implemented systems and changed source

Area	Files under Assets/KombiShift2	Result
Competition	Scripts/Traffic/RivalKombiAI.cs; RivalDrivingProfile.cs; KombiTrafficVehicle.cs	Demand/travel/congestion utility, hysteresis, physical wheels and torque, profiles, left lanes, clearance-gated passing, signal/zone caution, visible colours/activity
Passenger service	Scripts/Gameplay/PassengerTrip.cs; PassengerStation.cs; KombiShiftGameDirector.cs	Individual player trips, downstream destinations, bell/text requests, intermediate animation, delivery fares, one-shot missed penalty and recovery
Service quality	Scripts/Gameplay/KombiServiceQuality.cs	Separate XP and bounded passenger happiness, cooldowns, motion/collision/law reactions and smooth-service rewards
Condition	Scripts/Vehicle/KombiVehicleCondition.cs; RccHiaceTestBootstrap.cs	Impact speed/impulse filtering, bounded damage, throttle/brake adapter, paid stopped-service repair
Law	Scripts/Gameplay/KombiTrafficLaw.cs; KombiRoadRuleZone.cs; KombiRoadRules.cs	Configurable zones, speed grace, continuous offence incidents, stop/no-entry/direction/passing detection and offence records
World/stages	Scenes/KombiShift_FirstShift.unity; Editor/Upgrades/ShiftExpansionUpgrade.cs; Art/Generated; Scripts/Gameplay/KombiStageSettings.cs; KombiChallenge.cs; Assets/Resources	674.2 m -> 1,249.3 m itinerary (85.3% longer), 127 waypoints, seven stations, navigation for player/rivals/traffic, shared physical hump mesh, stage traffic/fare/profile differences
UI	Scripts/UI/KombiNavigationHud.cs; KombiMainMenu.cs	Existing design extended with requests, XP/happiness, condition, limit, fines, deliveries, rival proximity and repair help
Verification	Tests/Editor; Scripts/Development/ExpansionVerification.cs; RoadPhysicsVerification.cs; ExpansionVisual.cs; updated AssessmentVerification.cs; Editor/Validation/ExpansionValidation.cs; ExpansionBuild.cs	Repeatable component, UTF, physical scene, RCC hump, regression, render and build workflows

Docs/Assessment updates include current design/GDD, storyboard, event/state diagrams, traceability, readiness, test instructions, defence exercises and AI/asset declaration. Full changed-file inventory is in Docs/Validation/Expansion/changed-files.txt. Existing PDF, deck, source hand-in folders and packaged ZIP have not been silently replaced; they need regeneration after review.

Validation

Current evidence is under Docs/Validation/Expansion. Historical results retain their original scope/date.

- Rule/component fixtures: **25 passed, 0 failed**.

- Unity Test Framework EditMode: **15 passed, 0 failed** (utf-editmode.xml).

- Final actual-scene expansion integration: **28 passed, 0 failed** (integration.json). It observes real rival motion, shared-pool boarding, departure and then uses staged player positions for service/economy/reset checks.

- RCC hump physics: **5 passed, 0 failed** (road-physics.json). Safe staged crossing: 11.82 m progress, maximum observed vertical speed 1.34 m/s, health 100. Fast staged crossing: 48.92 m progress, health 96.79, upright. These are headless controlled observations, not a comfort study or a general no-launch guarantee.

- Initial expansion integration: 21 passed / 2 failed, retained as integration-first-run.json. Two >30 m waypoint gaps were densified. The repair fixture's insufficient funds were corrected by choosing damage affordable from its actual earned balance. Retest then passed 26/0; final departure assertions increased coverage to 28/0.

- Existing stage regressions, powertrain, visual inspection and export results: see final status appended below and pipeline.json. The fare assertions were updated for the deliberate boarding-to-delivery income change, with higher Rush fares. New results use a separate folder.

No human tester, feedback, contribution, textbook reading or member exercise has been fabricated. Rule fixtures and staged integration are not a human-driven complete shift. A successful export is not evidence of native hardware, browser or accessibility usability.

Build and execution status

New build output root: Builds/Expansion2026 in the parent workspace. Existing Builds/Assessment2026 releases are preserved. Each build writes platform/result/error/warning/size/time metadata under Docs/Validation/Expansion. See appended final status for actual outcomes. Current source must be used to import/run the expanded scene; previous binaries contain the earlier rules.

Known limitations and remaining definition-of-done work

1. The expanded map is a connected bidirectional itinerary using three existing road splines. It is not a general branching lane graph, and this work does not add an independently routed alternate street network or certify all road widths/intersections/endpoint turns. Blocked rivals wait and reconsider demand; they cannot plan an alternate road. No teleport recovery is used.

2. Passing requires side/rear/forward clearance and gradual physical steering. Controlled oncoming, overtaking return, endpoint-turn, intersection-priority and long deadlock scenarios need further deterministic integration coverage and human review. Current tests prove physical pickup/departure, not every manoeuvre's safety.

3. Conservative/competitive/aggressive differences are implemented, but stage targets and risk parameters are initial tuning. More aggression is not a measured balance result. Aggressive illegal passing is counted; a unified complete rival offence/economy log is not implemented.

4. Give-way has caution/speed behaviour; it is not a complete reliable yielding-offence detector. Signs use lightweight readable text rather than a full authored sign-art set. Zone extents and player passing classification require route playtesting.

5. Player passengers have individual intermediate trips. Rival passengers still follow their rival toward Mbare. Completed passenger records retain satisfaction, but there is no separately presented aggregate end-of-trip star rating.

6. Condition reduces power/acceleration and braking with controllability bounds. Suspension discomfort/damage is represented through hump consequences, not a complete per-wheel alignment/suspension wear simulator. Dedicated severe-damage smoke/impact VFX/engine-audio changes remain polish work.

7. New whole-game before/after CPU/GPU optimisation profiles and weak-machine results are not claimed. Existing historical traffic-query measurements are controlled editor evidence. Mesh/material reuse, nonalloc casts and half-second manifest sampling are implementation choices pending full gameplay profiling.

8. Legacy fixture assumptions were updated for the expanded stations, destination-aware progression, delivery fares and saved camera settings; only results listed as executed establish current validation. Complete cross-platform, resolution, sound, accessibility and long-session testing remains necessary.

Rubric evidence still requiring the team

Rubric area	Remaining evidence/work
Design/GDD 12	Human balance review; regenerate illustrated PDF from current design
Storyboard/prototype 8	Updated current storyboard is supplied; genuine early prototype feedback remains historical evidence to supply
Architecture 17 / core 13	Further manoeuvre, junction, long-shift and nonregression coverage; member understanding
AI/level/progression 10	Branching lane-graph scope gap, controlled passing/junction tests, human difficulty balance
UI/art/animation/audio 9	Human audibility/accessibility/legibility; lecturer-specific Animator expectations; remaining damage polish
Playtesting/iteration 9	Three authentic testers, observations/severity, three observation-linked fixes and same-task retests
Optimisation/deployment 7	Actual gameplay before/after profiling, native Windows/weak-machine/browser checks and fresh import
Report/source/evidence 5	Regenerate PDF/deck/submission snapshots, verify asset source permissions and final inventory
Panel defence 10	Every member personally practises and records actual understanding, contribution and live parameter exercises

Use EXPANSION_TEST_PLAN.md, EXPANSION_DEFENCE.md and the genuine tester CSVs. Resolve existing imported-asset source hand-in permission. Local commits and builds do not grant redistribution rights. No shared-history rewrite is used; the completed source/evidence commits are to be pushed to the requested branch.

Final platform and regression status

Executed check	Result	Scope
Expansion rule/component fixtures	25/0	Editor component/rule assertions
Unity Test Framework EditMode	15/0	Automated rule tests
Expansion scene integration	28/0	Physical rival pickup/departure plus staged player service
RCC hump physics	5/0	Safe/excessive staged crossings
Existing Morning regression	18/0	Editor staged integration
Existing Rush regression	18/0	Editor staged integration
Legacy passenger service	26/0	Expanded queues, physical rival collection, cancellation and delivery
Grounding after repairs	18/0	Dynamic physics with explicit controlled impacts
Existing powertrain	11/0	Dynamic wheel driving, gearing, RPM, governor and braking
Rendered menu/camera	22/0	Normal graphics-enabled Editor, saved PNGs
Linux standalone Morning / Rush	18/0 each	Native Linux process, headless staged verification
Windows executable Morning / Rush	18/0 each	Final repaired export under Wine, not native Windows hardware
WebGL Morning / Rush	18/0 each	Actual browser; before the final scene-only collision/spawn repairs

The first graphics-enabled *batch* capture produced no PNGs despite exit 0; it is not counted as a visual pass. A normal graphics window subsequently produced the saved screenshots. The first menu/camera run retained a failed zoom assertion that assumed a distance above 8 m regardless of

saved settings. The fixture now checks the actual requested delta and clamp without changing the user's preference; its retest passed 22/0. These are technical fixture findings, not human playtest feedback.

Export	Result	Errors	Warnings	Size reported by Unity
StandaloneLinux64	Succeeded	0	59	394,778,802 bytes
StandaloneWindows64	Succeeded	0	59	395,244,885 bytes
WebGL	Succeeded	0	64	200,190,952 bytes

Release runtime systems originate in commit 593b134; final scene repairs and rival spawn changes are included through aaa41b5. Refreshed exports were built after those scene changes. Scene GUIDs and vendor dependencies were retained. The parent workspace launcher now prefers the validated expansion Linux binary, with the earlier release as fallback. Parent README/launcher edits are workspace changes outside the nested game repository. Push to the requested branch is authorised; the commit history records the reviewable changes.

Browser first-load scene import was slow, and simultaneous Editor/browser work caused memory pressure. The project has runInBackground disabled; a hidden browser can pause a verification run. These constraints must be considered when reproducing browser timing. Audio metadata loading messages were observed during WebGL loading; real audibility is still a human check.

Collision regression found by the legacy service suite

The adapted legacy suite initially passed 23 checks and failed three linked rival/queue checks. The rival stopped 11 m before Arterial Junction, grounded on all four wheels with zero motor torque and active brakes. Its sensor reported tree1Plane_001 589 2.16 m ahead. The rendered road was visibly clear: an imported alpha-cutout tree-plane collider projected into the road corridor. This was a real newly activated-road collision defect, not a passenger-reservation failure.

ShiftExpansionUpgrade.RepairRouteTreeColliders disables affected scene-instance billboard mesh colliders and adds narrow trunk capsules; imported art, asset files and scene GUIDs stay intact. Sixteen oversized plane colliders near the route were replaced. The rival then physically reached the second stop, boarded all six people and departed. The legacy service retest passed **26/0**, including cancellation/return, shared queue depletion, seat allocation, minimap layout and final delivery. Its initial failed JSON and physical sensor logs are retained. These are engineering observations, not authentic tester feedback.

All three platform exports were refreshed after the scene repairs; final-check-pipeline.json records their successful completion. Earlier platform execution records remain separately scoped to the pre-repair exports; final native Linux smoke results identify the refreshed release.

Final repaired-scene verification

- Legacy service: **26 passed / 0 failed**.

- Grounding: **18 passed / 0 failed**, including dynamic boarding, steering, reverse, kerb/traffic-body impacts, wheel contact and no recovery teleport.

- Expansion integration: **28 passed / 0 failed** after the final scene changes.

- Linux, Windows and WebGL refreshed exports all succeeded; final build summaries and content identities identify their exact outputs.

The initial grounding run passed 17/18; its player hit a waiting rival before the steering exercise. Delayed rivals now start behind the player, preserving the intended clear head start. The controlled handling fixture isolates background vehicles and measures peak heading change through opposing turns, while retaining explicit kerb and traffic-body collision tests. Its retest passed 18/18. The route now has 127 points and measures 1,249.3 m versus 674.2 m originally (85.3% longer).

Current report PDF: output/pdf/KombiShift2_Expansion_Report_2026-11-10.pdf. Earlier illustrated report, deck and submission ZIP/source copies remain historical snapshots; update the final hand-in package and deck from this current source.

The refreshed Linux export passed **18/0 in each stage** (native-linux-final-Morning.json, native-linux-final-RushHour.json). The current dated PDF is provided alongside this Markdown source; regenerate it with Tools/BuildExpansionReport.py.

The refreshed Windows export also passed **18/0 per stage under Wine** (wine-final-Morning.json, wine-final-RushHour.json). WebGL feature integration passed before the final scene-only changes; its refreshed export succeeded, but that last browser run was not repeated. Native Windows hardware and weakest-device checks remain outstanding.

4 / Panel practice

These are exercises to perform, not completed member achievements.

- Explain why a busy distant station can beat a nearby small queue. Change RivalDrivingProfile.passengerValue or switchAdvantage and observe a real target decision. Explain reservation ownership, physical arrival and cooldown.

- Show WheelCollider steering/motor/brake torque, forward/rear/side pass clearance and gradual lane offset. Explain why the current itinerary is not a general lane graph and how endpoint turns can fail.

- Trace a passenger: station coroutine -> actual seat callback -> PassengerTrip -> request -> alighting animation -> completed fare/XP. Show a missed trip and recovery. Character state differs from trip state.

- Change missedStopPenalty or missedStopXP on the director. Demonstrate the one-shot flag and reaction cooldown. Explain why current happiness is separate from cumulative XP.

- Explain relative collision speed plus impulse/mass filtering, incident damage cap and bounded condition multipliers. Show RCC remains the player controller and vendor code is untouched.

- Change a speed-limit zone or grace period. Explain zone contacts with multiple colliders, one continuous speeding incident and separate fine/XP/happiness effects. Distinguish game fines from real law.

- Change MorningShift/RushHour configuration. Explain shared logic, traffic density, rival risk and higher Rush fares. Do not claim balancing from testers until real results exist.

- Read test evidence: identify initial failures, fixes and retests. Distinguish staged positions from driven navigation, editor rule checks from builds, exports from native execution and automated checks from authentic human sessions.

- Demonstrate an actual before/after gameplay profile on a target machine. Explain any unmeasured GPU or hardware claim as a gap, not a result.

Every member must personally practise and record real reviewed/modified files and explanations in CONTRIBUTIONS.csv. AI assistance and imported assets must be disclosed accurately. Existing PDF/slides describe the earlier release until updated.

Evidence and source locations

Structured results, scene audits, source hashes and build identities: Docs/Validation/Expansion. Changed files: changed-files.txt. Reproduction and authentic tester tasks: Docs/Assessment/EXPANSION_TEST_PLAN.md. Existing PDF/deck/submission snapshots are preserved as earlier-version material.

AI and assets

OpenAI Codex assisted implementation, technical verification and report preparation at Jayden's request. Imported Unity/RCC/city/vehicle/character assets remain attributed to their existing sources. Members must record actual understanding and contributions. Authentic feedback, member exercises, assigned-book reading and source redistribution permission are not asserted by this report.